

Custom AI Data Analysis and Software Opportunity with the National Geographic Exploration Technology Lab

Collaboration Opportunity:

The National Geographic Exploration Technology Lab supports Explorers working with large datasets who could benefit from AI-driven data processing or automation to enhance their bespoke data analysis efforts.

This opportunity is for Explorers with collected data, such as image, video and geospatial, to work with a team of computer science students over an intensive 14-week collaboration period during which the students will perform agentic AI data analysis or develop specialized utility software to help Explorers and their teams derive new insights.

Previous projects include the development of software to detect amphibians in 15,000 hours of audio; detect and classify fish in underwater images; detect whale calls in passive acoustic monitoring data; and build software tools to help review field recordings for musical compositions. The projects can be AI-based (detection and classification) or support the creation of bespoke software interfaces or utility software to aid Explorers and their teams in reviewing collected data.

This collaboration will run from September to December 2026. Selected Explorers will collaborate directly with the student teams. The program will produce a proof-of-concept solution tailored to each project, and Explorers are expected to actively engage three to four times a week throughout the collaboration.

This opportunity is open to any active project led by an Explorer and is not limited to work funded by a National Geographic Society grant. You do not need to have an active NGS grant to apply. Questions about the opportunity can be directed to Kasie Coccaro at kcoccaro@ngs.org. This opportunity closes on September 18, 2026. Selected finalists will be contacted to discuss their projects. At that time, we may ask for sample data to be shared for review. Final selection will be made prior to September 30, 2026.

Eligibility and other details

- Applicants must be National Geographic Explorers to apply.
- While preferable, the proposed project does not need to be related to an NGS grant. This opportunity is open to any Explorer who can demonstrate a well-defined data analytics challenge.
- Explorers should demonstrate the ability for active participation and review of deliverables, e.g., quickly reviewing analysis for accuracy, conducting multi-person testing of software, or thoughtfully applying achievable directional input. This project requires deep human-in-the-loop participation to achieve results. In addition, a key value of this opportunity to the students is the chance to connect with the selected Explorers throughout the project period. If you have limited communication during this period, you are still welcome to apply. Please note your full availability on your application so we can consider you for a future engagement.
- The Explorer must be available for a weekly call (up to 60 minutes) with the student team to provide guidance and feedback throughout the initial project period (September to December 2026). A research assistant or project team member may join the calls, but the Explorer is expected to attend at least 70% of the sessions. For emails and other communication, a 12-hour turnaround period is preferred.
- This is a 12-month total collaboration period, with the first four months (September to December 2026) dedicated to the data collaboration processing. The following eight months will be undertaken independently by the Explorer, with the expectation of a final, impact-driven output if the initial data processing yields value.
- Applicants will be required to sign a collaboration agreement.

Project Specifications and Considerations:

- Applicants should be mindful that this is a limited-time engagement for undergraduate and graduate engineering students and should scope project deliverables accordingly. Therefore, Explorers should propose a minimum viable product (MVP), not a large-scale or production-ready tool.
- Applications that are tailored for short-term impact or results within an eight-month timeframe following the completion of the student collaboration are preferred. This allows us to narrow the technical problem statement to an achievable result.
- Upon completion of the project and demonstration of manuscript-worthy findings (based on consensus by all parties), funds to offset publication costs may be made available on a limited, non-guaranteed basis.

Data Requirements:

- The Explorer has a clear understanding of the methods, ethics, and necessary approvals (e.g., permits or legal agreements) for sharing relevant data. The Explorer bears all responsibility for ensuring that data is legally and ethically obtained and able to be transferred to a third party (U.S.-based).
- The Explorer retains all rights to the data and final analysis. Additional requirements will be outlined in a collaboration agreement that must be signed upon project acceptance.
- This opportunity is structured for large amounts of data, e.g., hundreds of hours of audio, video, and hundreds or thousands of photos. In the application, please specify the amount of data and how computer-aided analysis is needed to achieve your goals.
- The Explorer will deliver all assets before the start of the project. The ability to coordinate the timely sharing of the data is a must. Applications must demonstrate the ability to quickly provide the data sets for analysis. Before the project begins, institutional and/or governmental permissions must be obtained to share the data. Successful applications will be able to provide initial data (including annotations/labeled data) in early October 2026 and continue to provide any additional data needed throughout the engagement. For this to be achieved, Explorer data will need to be previously collected and stored in a systematic structure on a cloud-based system. If currently stored locally, a cloud transfer will be made available for upload within a week of acceptance.
- The Explorer retains ownership of all supplied raw/original data, in addition to any manipulated and derived data and insights (e.g., annotations, bounding boxes and labels), documentation and metadata. For clarity, the underlying software code is not owned by the Explorer. Software front-end systems and data analysis frameworks will be provided free of charge for the Explorer's private use on a limited-term basis for the agreed-upon purpose. Additional details will be outlined in the collaboration agreement.

End Product:

All end products will be discussed and documented during initial project scoping. As an example, at the end of the engagement, the Explorer could receive the MVP (e.g., online software, Python scripts, machine-learning models and the output data from annotation efforts). All software and code will be delivered with instructions for use. The Explorers retain the right to use data outputs and inputs as they see fit, as long as it falls under the initial project scope. One or more of the weekly calls will be devoted to ensuring the Explorer can use the product and understand the underlying technology.

How to Apply

Click on the following link to apply: [Application Form](#)

Please note, this opportunity is not an endorsement of products or services, and the Society is providing the software and services on an "as-is" basis, meaning that the Society is not responsible for the specs, inner workings or even whether the product is suitable for a particular application. You are responsible for assessing the trustworthiness of the software. You are also responsible for deciding what, if any, personal data you share or store.